3/26/11

SEQUENCES

Def: Let X be a set. A "sequence" in X is a function $x: \mathbb{N} \longrightarrow X$ i.e., an infinite collection of points $x_1, x_2, x_3, \ldots$ $(x_n = x(n))$

Def: Let (X, T) be a topological space and let $x \in X$. We say that the sequence $(x_n)_{n \in \mathbb{N}}$ "converges to"/"tends to" x and write $x_n \longrightarrow x$, if for every neighborhood U of x, $\exists N \in \mathbb{N}$ s.t. $x_n \in U$ $\forall n \geq N$.

Eg: 1. In $\mathbb{R}$ with the std. topology.

$$1/n \longrightarrow 0,$$

$$1/n^{1/\varepsilon} \to 0 \ \forall \varepsilon > 0$$

$$(1 + 1/n)^n \longrightarrow e$$

2. Sequences in general spaces may not converge at all, or may converge to more than one point.

* In $\mathbb{R}$ with the std topology, the sequence $x_n = n$ does not converge.
* In the indiscrete topology $T = \{\phi, X\}$ on any set X, every sequence converges to every point.
* Line with doubled origin: Consider the non Hausdorff space $\mathbb{R} \setminus \{0\} \cup \{0_+, 0_-\}$ introduced in a previous lecture.

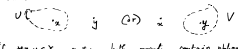
  Here, $1/n \longrightarrow 0_+, 0_-$.

Separation Axioms

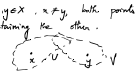
These are properties that measure how "separated" the closed sets of a topological space are.

A topological space (X, T) is said to be:

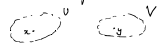
* T_0, if $\forall x, y \in X$, $x \neq y$, $\exists$ an open nbhood of at least one of the points that does not contain the other.



* T_1: if $\forall x, y \in X$, $x \neq y$, both points contain nbhoods not containing the other.



* T_2, if $\forall x, y \in X$, $x \neq y$, both points have nbhoods that are disjoint.



Eg:

1. A topological space that is T_0, but not T_1:

   $\mathbb{R}$, with a topology where the open sets are of the form (a, ∞)

   Given $x \neq y$ (WLOG $x < y$).

   $\left(\frac{x+y}{2}, \infty\right)$ contains y but not x.

   However it is not possible to find an open set containing x but not y.

2. Space that is T_1, but not T_2

   Cofinite topology on $\mathbb{R}$:

   No two open sets can be disjoint so this space is not T_2.

   Given $x \neq y$ in $\mathbb{R}$, $\mathbb{R} \setminus \{x\}$ is an open nbhood of y that does not contain x, and $\mathbb{R} \setminus \{y\}$ is an open nbhood of x not containing y.

Remark: $T_2 \Rightarrow T_0 \Rightarrow T_1$ [There are T_n spaces for $n \geq 3$, but we will discuss these later]

Sequences and limit points

Fix a T_1-space (X, T) and let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X, and suppose $x \in X$.

Prop: For any $A \subset X$, $x \in L(A) \Leftrightarrow$ every nbhood of x contains infinitely many points of A.

Proof:

Prop: Let $A = \{x_1, x_2, x_3, \ldots\}$. Then $x \in L(A) \Leftrightarrow \exists$ a subsequence $x_{n_k} \longrightarrow x$.

Proof:

Prop: If X is T_2 (Hausdorff), any sequence of points converges to at most one point in X.

Proof: